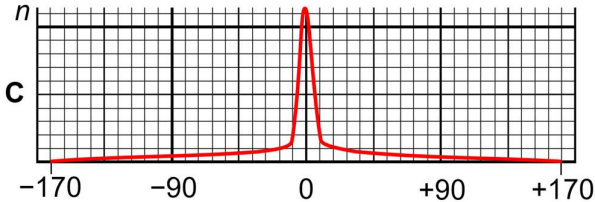


28	Answer: A																								
From notes, - Majority of α-particles went straight through or were deviated by small angles of less than 10°. - A small proportion (about 1 in 8000) of the α-particles were deflected through large angles of more than 90° or came straight back. A typical graph of N against θ should look like this, indicating a significant drop from the straight-through position to deviation of 10°:  It is hence useful to make sense of the numbers at key points of the given graphs <table><tr><th></th><th colspan="2">N</th><th></th></tr><tr><th></th><th>0° (Straight-through)</th><th>10°</th><th>Remarks</th></tr><tr><td>Option A</td><td>10^8</td><td>10^5</td><td>Decrease to 1/1000 of straight through</td></tr><tr><td>Option B</td><td>0.7×10^9</td><td>0.35×10^9</td><td>Decrease by 50%</td></tr><tr><td>Option C</td><td>0.7×10^9</td><td>0.68×10^9</td><td>Decrease by less than 10 times</td></tr><tr><td>Option D</td><td>10^8</td><td>$10^{7.8}$</td><td>Decrease by less than 10 times</td></tr></table> Option A shows the drastic drop in N from the straight through to 10° deviation position. This also implies that beyond 10° deviation, there is only a small fraction of particles with those angular deflection.			N				0° (Straight-through)	10°	Remarks	Option A	10^8	10^5	Decrease to 1/1000 of straight through	Option B	0.7×10^9	0.35×10^9	Decrease by 50%	Option C	0.7×10^9	0.68×10^9	Decrease by less than 10 times	Option D	10^8	$10^{7.8}$	Decrease by less than 10 times
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